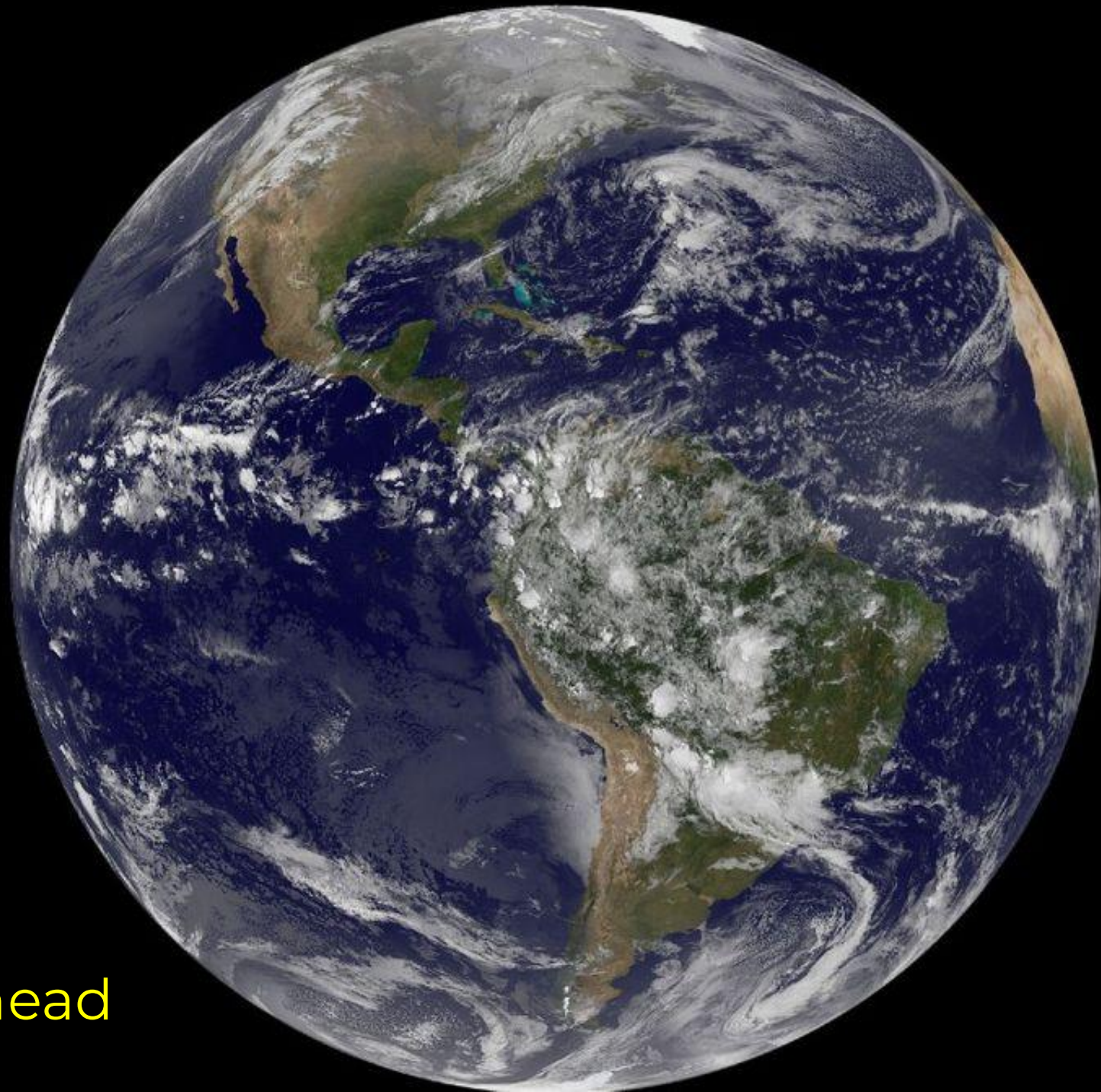


Deep Learning

Challenges ahead



CSCI 1470

Eric Ewing

Thursday,
12/4/25

Logistics

DL Day, December 11th (a week from today)

- 2 timeslots throughout the day, if your full group can only make part of a timeslot, I'll prioritize viewing your poster during that period of time
- Will take place in CIT 3rd floor atrium
- You just need to bring your poster
- Poster printer in CIT (4th floor printer room) has limited hours this year 8:30-4:30.
- We recommend aiming to print your poster on December 9th or the 10th. Do not wait until the 11th!
- Each poster takes ~10-15 minutes to print

Challenges in RL and Robotics

- Simulation environment and real world won't match perfectly (Sim2Real Gap)
 - Hard to collect enough data in the real world
 - Impossible to simulate physics perfectly
- No guarantees of safe policies
 - If you follow a learned and cause an accident, that's very expensive
- Sparse/Delayed rewards
 - It is challenging for a robot to know if it is doing well until a task is complete
- Partial Observability in the real world
 - Robots do not have access to the entire world state, just what they can observe with their sensors.

Why don't we see more RL in more deployed robots?



Why don't we see more RL in deployed robots?

Industrial robots work in *very controlled* environments

Deep Learning is not the answer to every problem

We already know optimal-control algorithms for certain types of problems, Deep RL cannot be better than optimal solutions...

The strength of Deep Learning is its ability to handle uncertainty and generalize to new data/environments

But there's lots of problems left



How could we create
generally intelligent robots?



General Intelligence

What properties do we want from a generally intelligent robot?

- 1. Adapt to new environments and tasks quickly**
2. Goal alignment and value learning
3. Work with multi-modal data
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6. Explainability and Interpretability

Adapt After Training: Continual Learning

What do you do when you encounter new data?

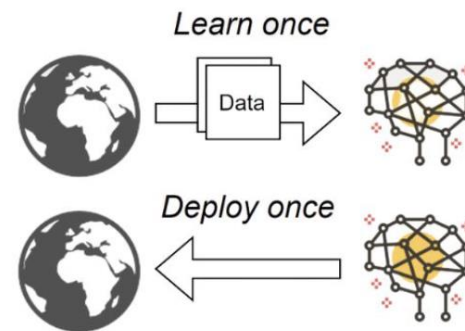
Keep trying to update your model...

2 things may go wrong:

Catastrophic Forgetting: The network no longer knows how to complete a task it once knew

Loss of Plasticity: The network can no longer learn and adapt to new tasks

Static ML



Adaptive ML

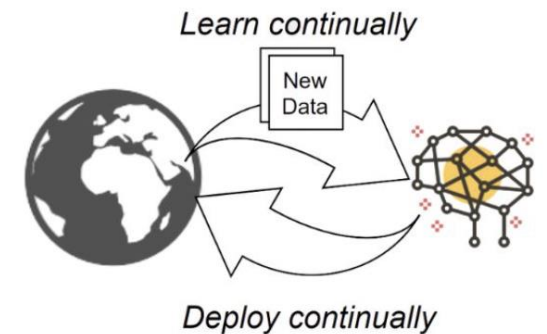
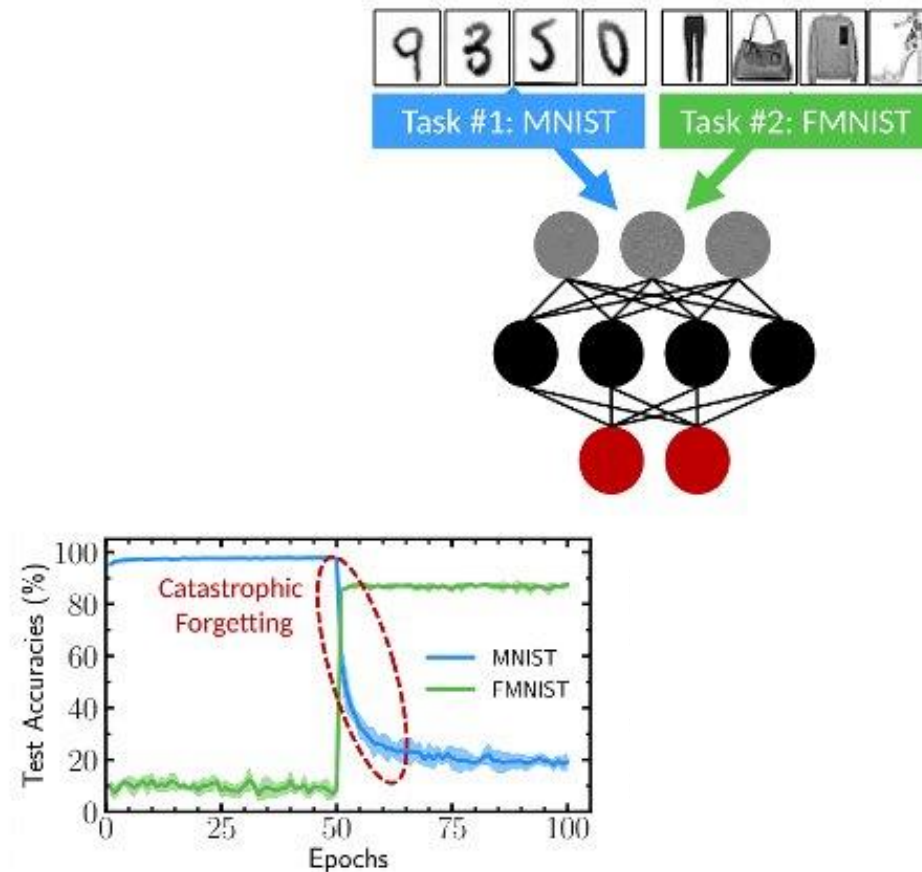


Image source: <https://imerit.net/blog/a-complete-introduction-to-continual-learning/>

Catastrophic Forgetting

Train network on MNIST,
then switch to FMNIST
(separate outputs)

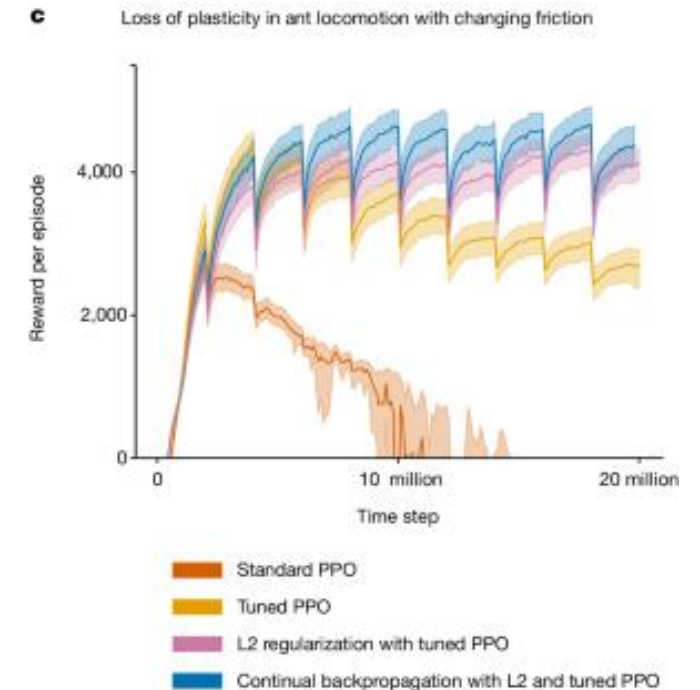
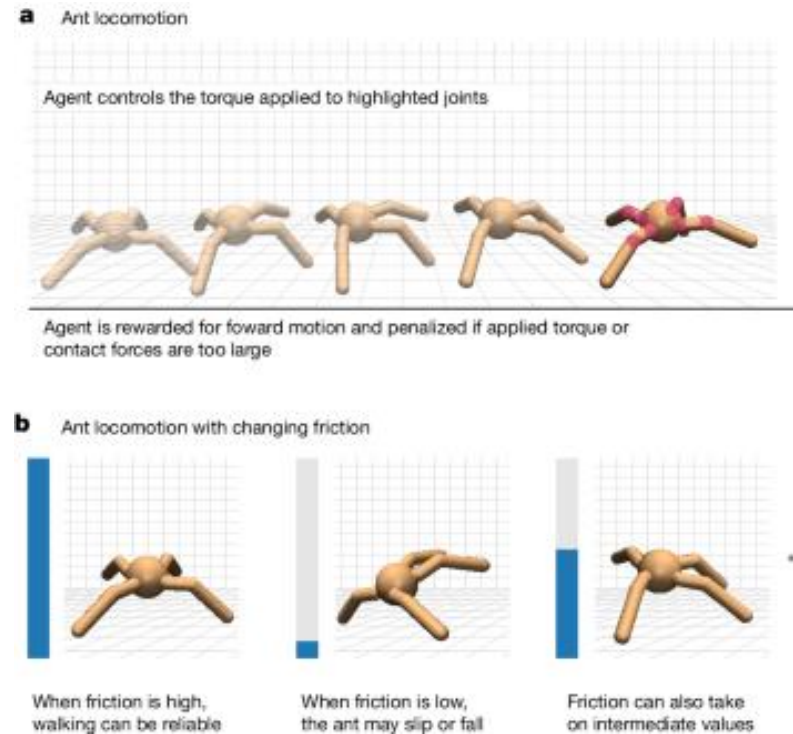
Ideally, our networks
would remember how to
complete the MNIST task



Loss Of Plasticity

Catastrophic forgetting is a problem whenever the task switches

But even worse... the network may not learn to complete new tasks



Continual Backprop

Calculate *utility* of each neuron in network

Reinitialize neurons that do not contribute to the output

Continue to run SGD on dataset

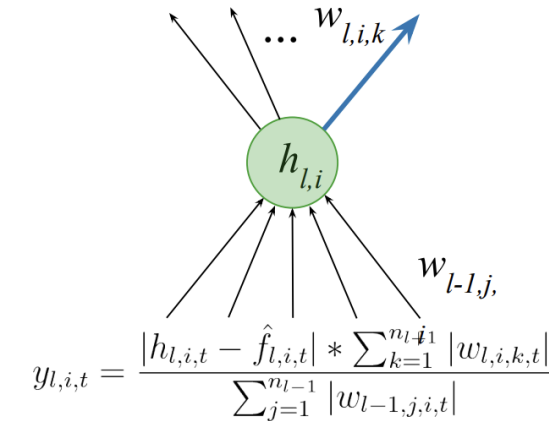
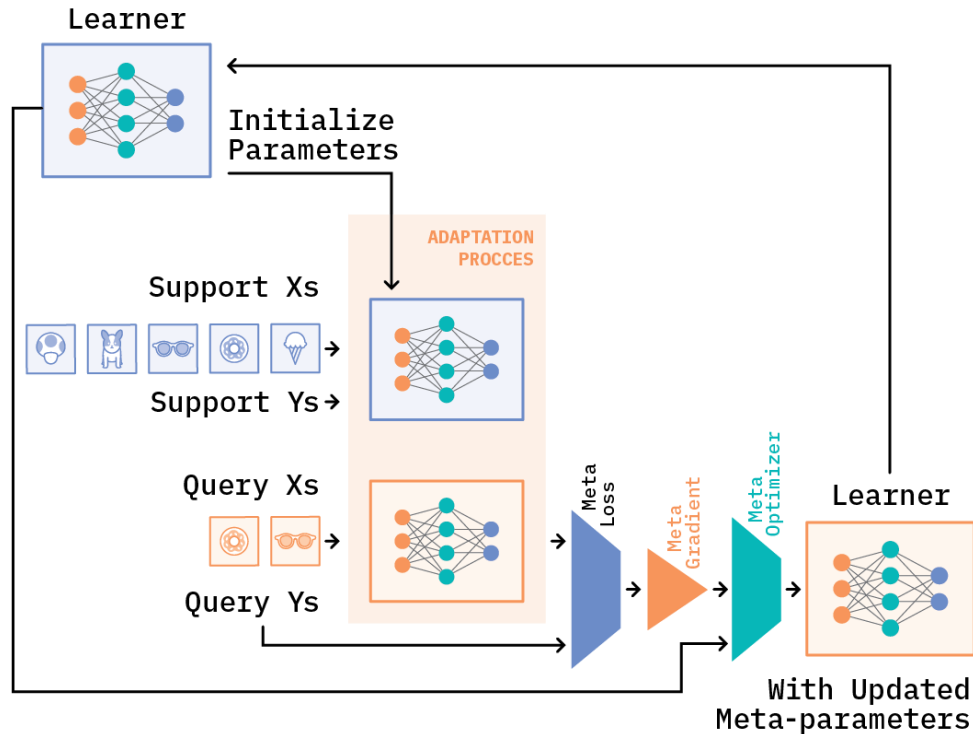


Figure 4: A feature/hidden-unit in a network. The utility of a feature at time t is the product of its contribution utility and its adaptation utility. Adaptation utility is the inverse of the sum of the magnitude of the incoming weights. And, contribution utility is the product of the magnitude of the outgoing weights and feature activation ($h_{l,i}$) minus its average ($\hat{f}_{l,i}$). $\hat{f}_{l,i}$ is a running average of $h_{l,i}$.

Adapting to New Tasks: Meta-Learning

Train a model that can adapt **quickly** to new tasks



Algorithm 1 Model-Agnostic Meta-Learning

Require: $p(\mathcal{T})$: distribution over tasks

Require: α, β : step size hyperparameters

- 1: randomly initialize θ
 - 2: **while** not done **do**
 - 3: Sample batch of tasks $\mathcal{T}_i \sim p(\mathcal{T})$
 - 4: **for all** $\mathcal{T}_i$ **do**
 - 5: Evaluate $\nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(f_{\theta})$ with respect to K examples
 - 6: Compute adapted parameters with gradient descent: $\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(f_{\theta})$
 - 7: **end for** **Note:** the meta-update is using different set of data.
 - 8: Update $\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'_i})$
 - 9: **end while**
-

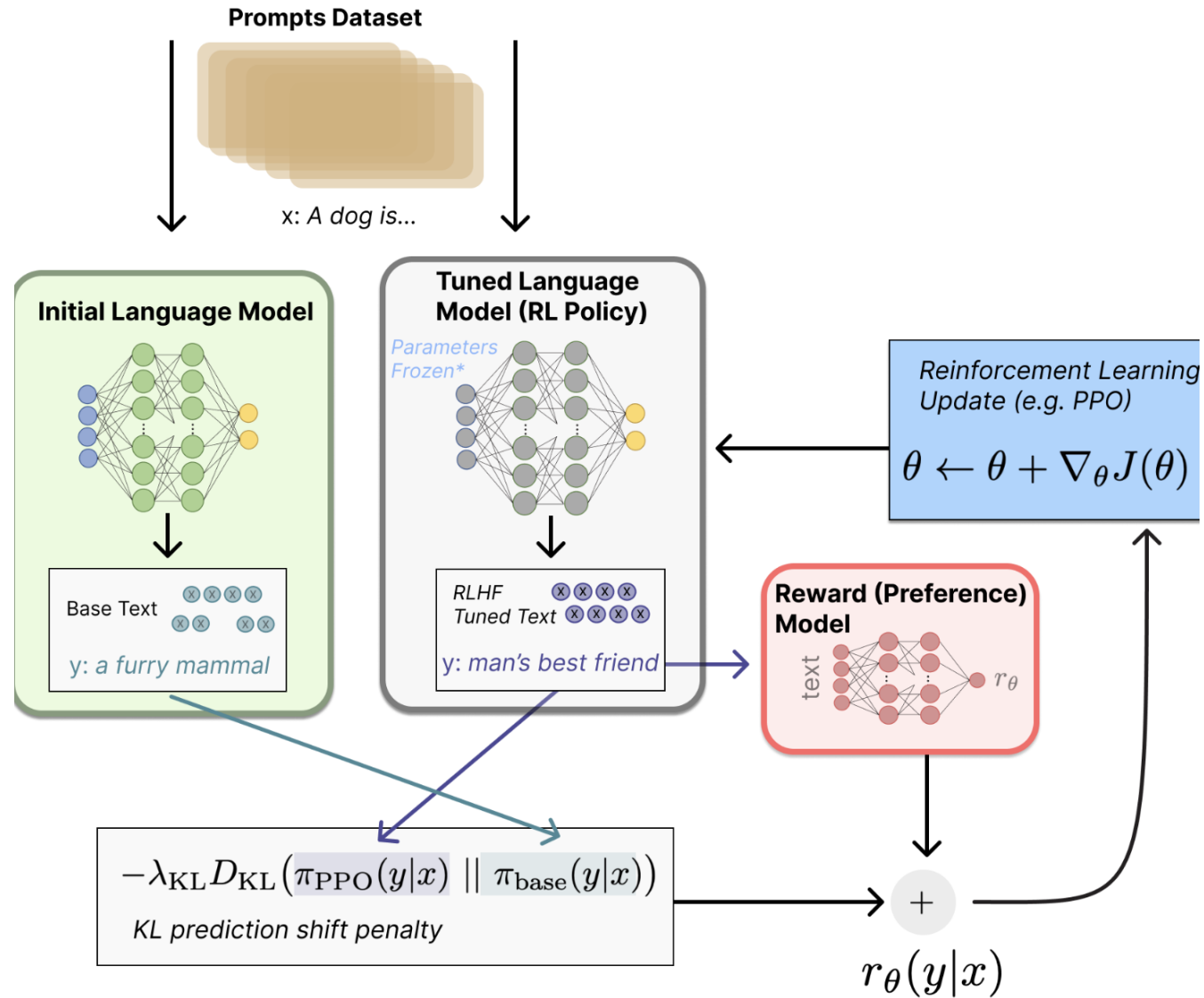
Model Agnostic Meta-Learning (MAML)

General Intelligence

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RLHF is a way to perform alignment



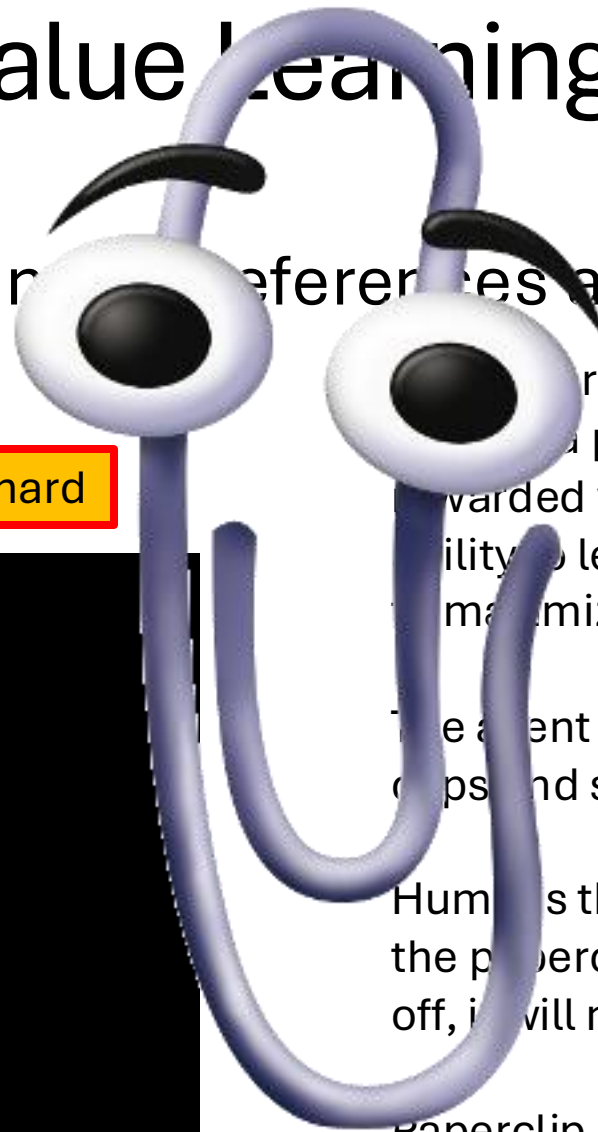
Alignment and Value Learning

How can robots learn human preferences and what we want them to do?

Specifying reward functions is hard



Positive reward for surviving, negative reward for losing



paperclip parable:

A paper clip factory and train an agent that is rewarded when it produces a paper clip. We give it the ability to learn even better strategies. The agent wants to maximize reward.

The agent needs to secure more resources for paper clips and starts strip mining.

Humans think strip mining is bad, and want to turn off the paperclip AI. The paperclip AI knows if it is turned off, it will no longer get rewards.

Paperclip AI wipes out humanity so that it can continue to make paperclips.

Learning Human Preferences

Given expert demonstration data, how can we learn to imitate the expert policy?



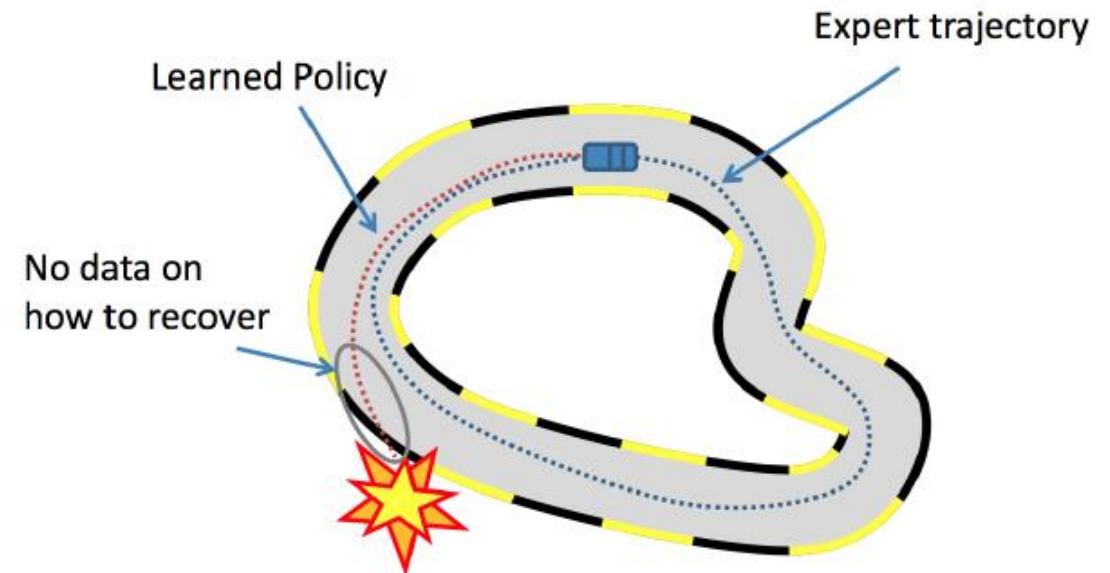
Imitation Learning

Behavior Cloning seeks to *imitate* the expert policy

Given a dataset of (state, action) pairs (i.e., trajectories), use supervised learning to learn a policy $\pi_{\theta}(s)$

Potential Issues:

What happens when we encounter a state we don't have data for?



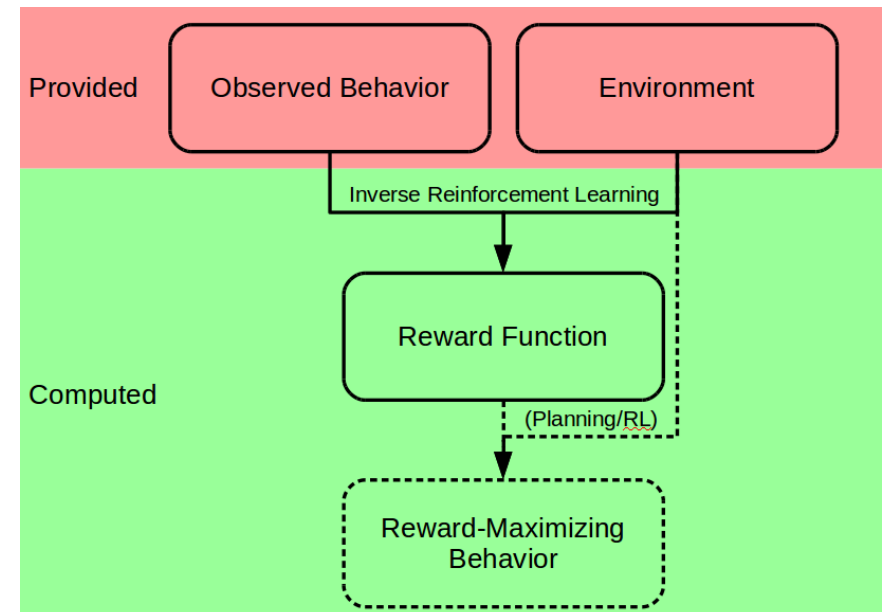
Inverse Reinforcement Learning

IRL seeks to learn the reward function of the expert

Given a dataset of (state, action) pairs (i.e., trajectories), learn a reward model. Then, use RL with the learned reward model to learn a policy

Potential Issues:

There is more than one reward function that could reproduce the given trajectories. How do we decide what the correct reward function is?



Inverse Reinforcement Learning

IRL is an **underspecified** problem

An infinite number of reward functions lead to the set of trajectories in our dataset, so which one are we trying to find?

Open Question: How do we find the “best” reward function given data?
What does it mean to be the “best” reward function?

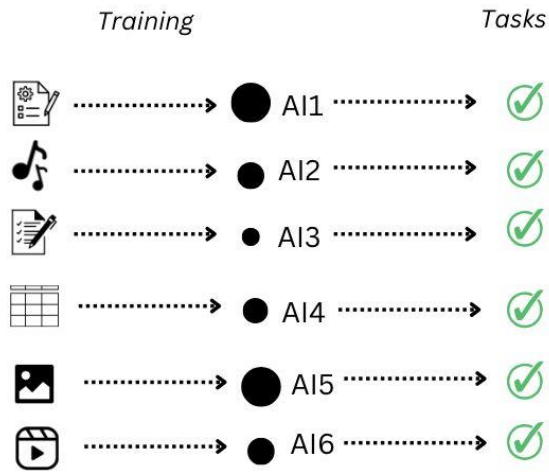
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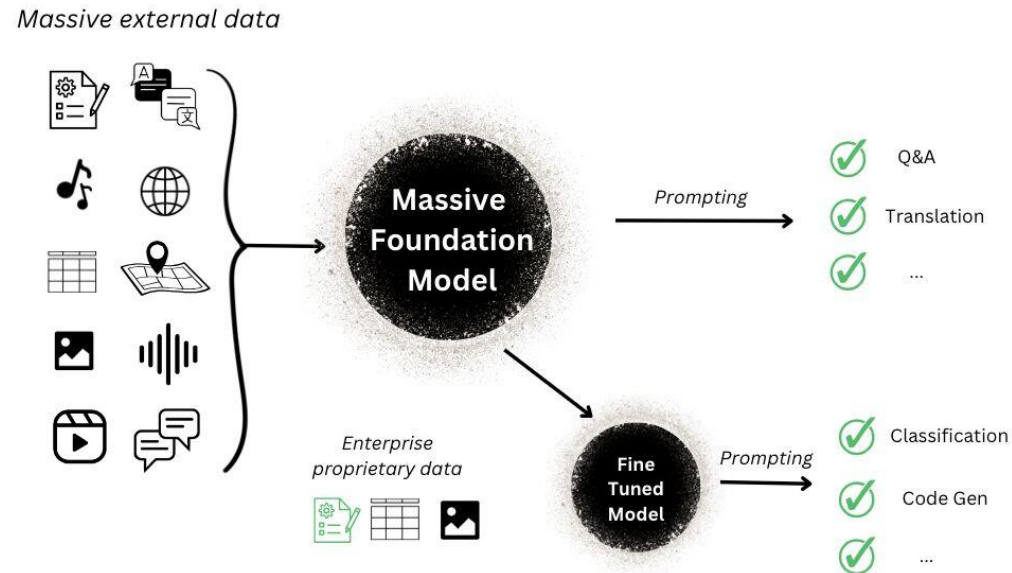
Working with Multi-Modal Data

Traditional ML



- Individual siloed models
- Require task-specific training
- Lots of human supervised training

Foundation Models



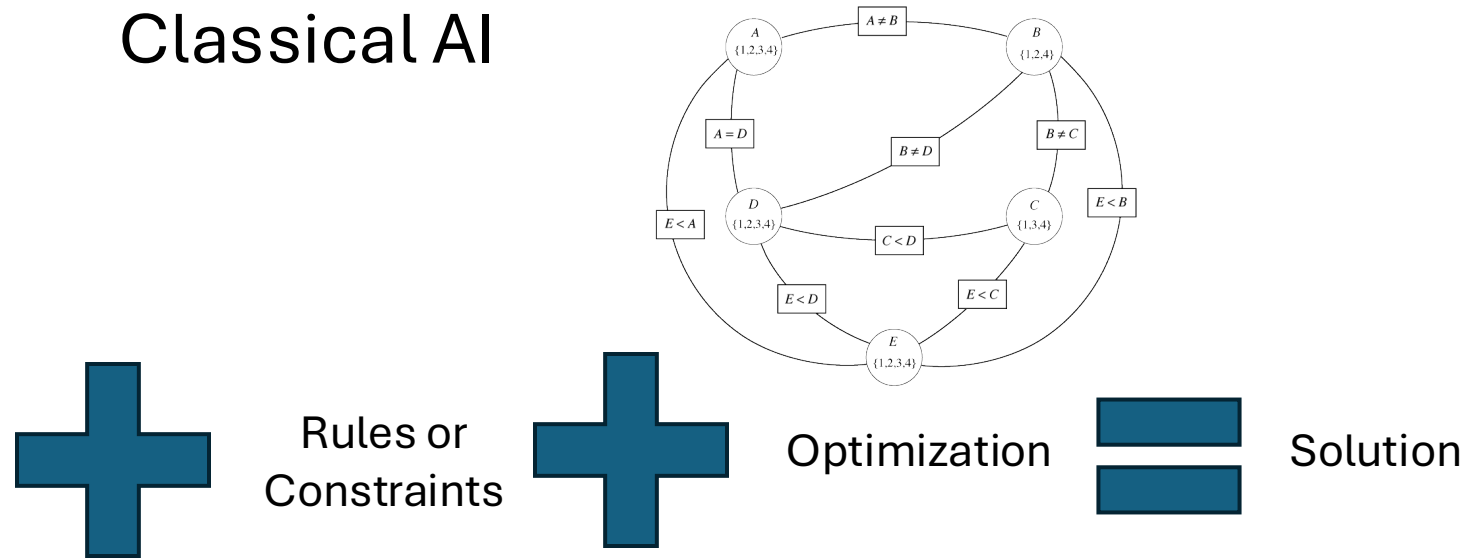
- Massive multi-tasking model
- Adaptable with little or no training
- Pre-trained unsupervised learning

Representations Aren't Free

8			4	6			7
					4		
	1				6	5	
5		9		3	7	8	
				7			
	4	8		2	1		3
	5	2				9	
		1					
3			9	2			5

This is represented as a 2D matrix

Classical AI



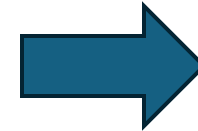
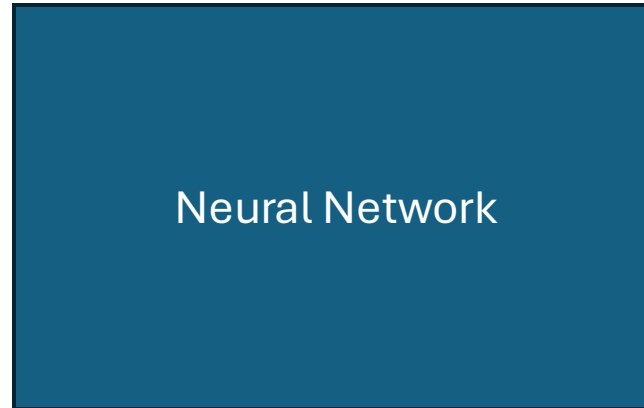
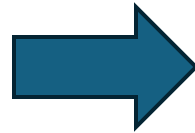
These are hardcoded

This is very general, but requires a very specific representation

Representations Aren't Free

Naïvely applying Deep Learning

8			4		6			7
						4		
	1					6	5	
5		9		3		7	8	
				7				
	4	8		2		1		3
	5	2					9	
		1						
3			9		2			5

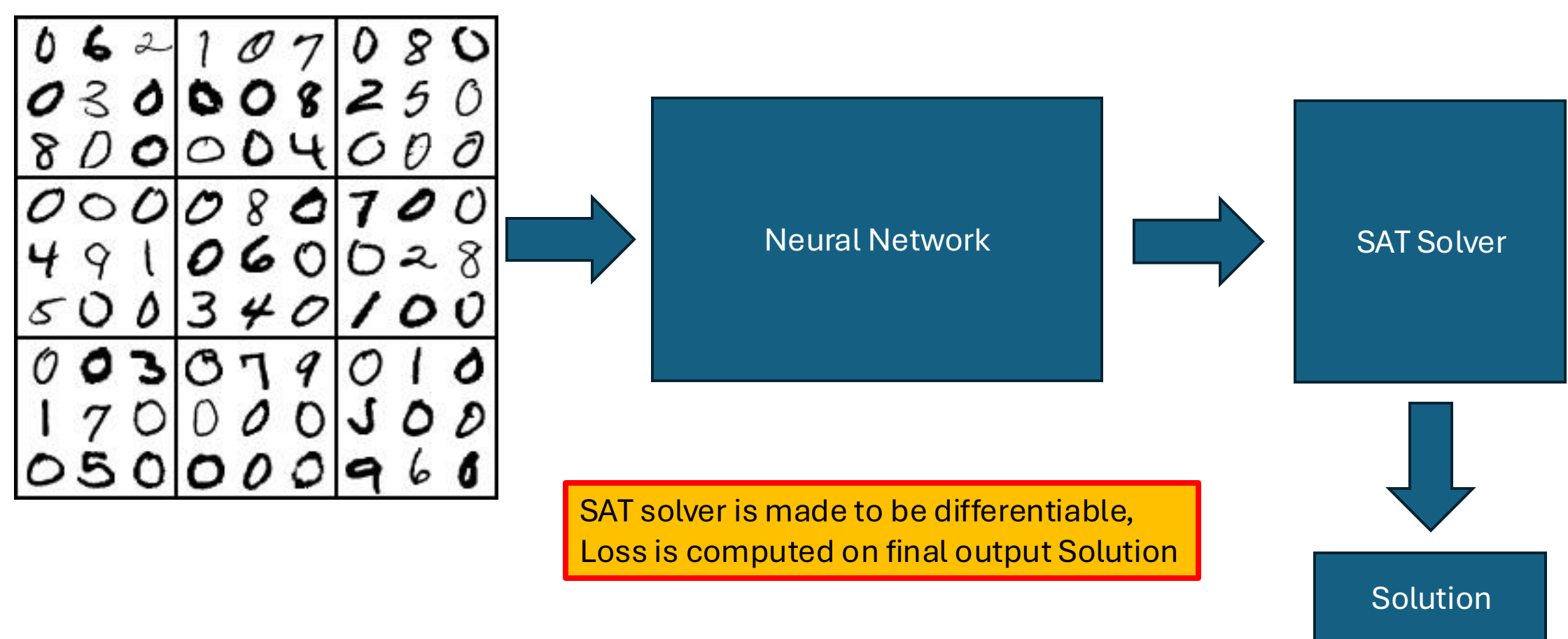


Solution

How much harder is the image-based Sudoku?

What if your network doesn't start with the ability to recognize digits (i.e., wasn't pre-trained on MNIST).

Differentiable Optimization Functions as Layers



General Intelligence

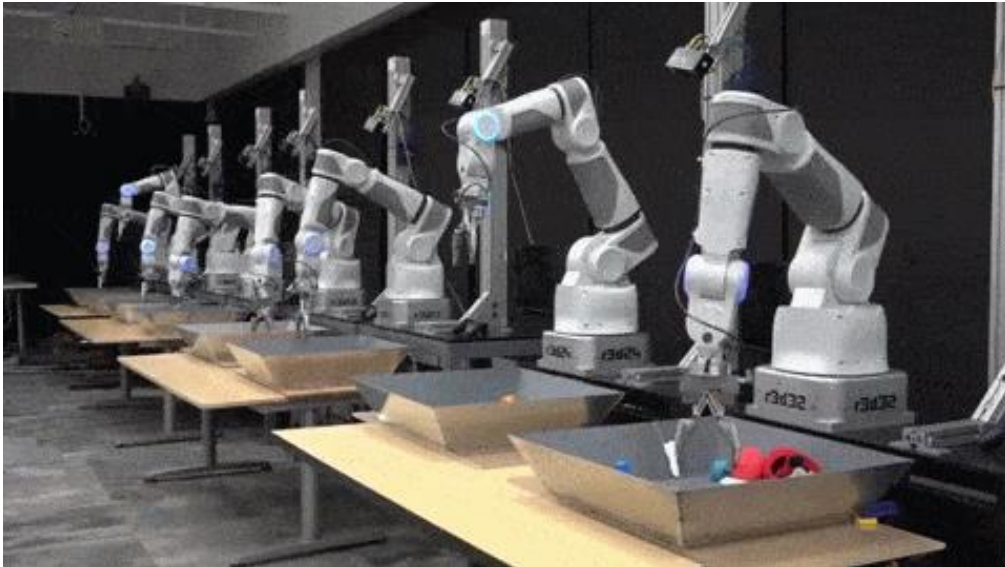
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Safe Exploration

Two issues with using the real world to collect trajectories for RL:

1. The real world is too slow... (simulations are fast to run)
2. Before Robots learn a “good” policy, they may take dangerous actions



Google Research

How do we still encourage safe exploration?

Failure Recovery?

Easiest way: Train failure recovery policy through a separate process (i.e., reward standing up). Learn when to switch policies

Harder: Learn a single general policy to optimize desired objective



General Intelligence

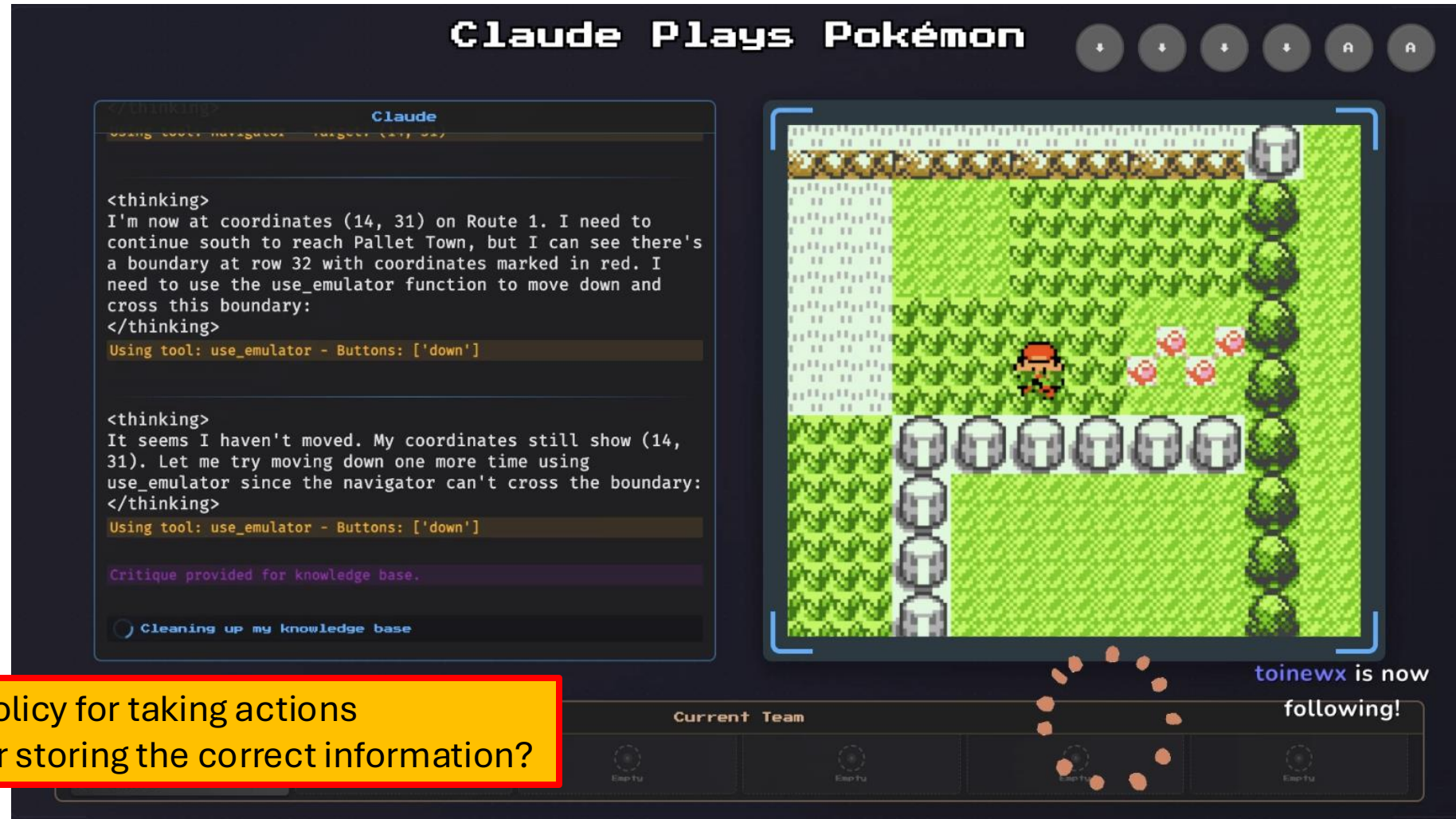
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Long-Term Memory

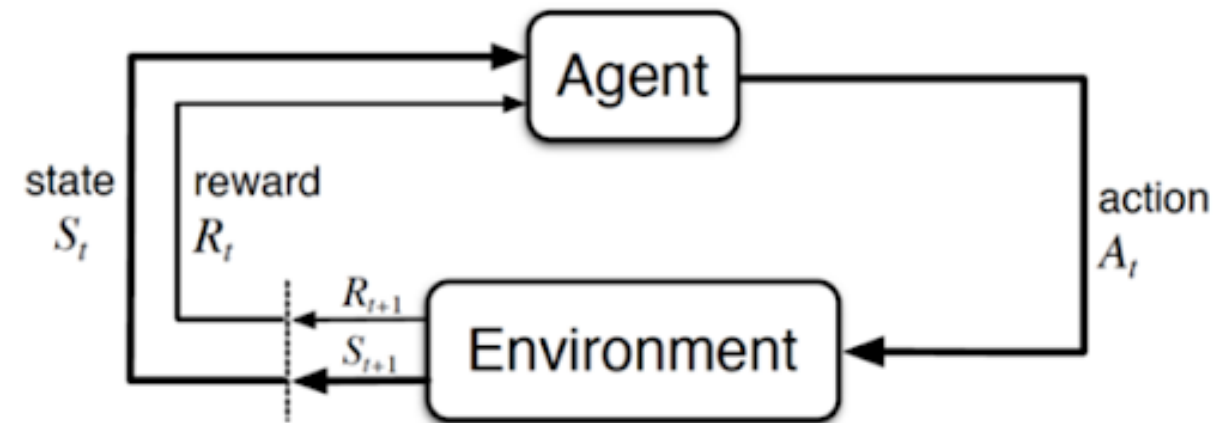
1. What should be stored in long term memory?
2. How should it be stored?
3. How can it be accessed?

Reward signal helps to learn a policy for taking actions
How can we reward our agent for storing the correct information?



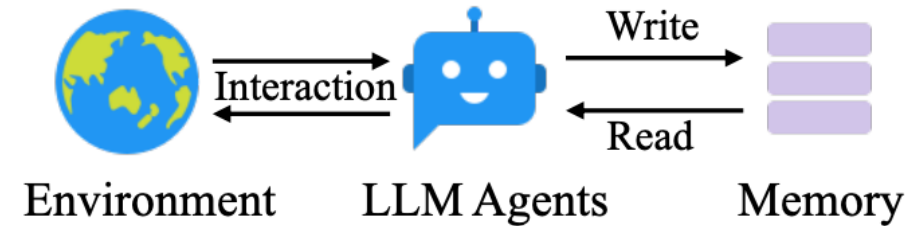
Claude plays pokemon

Imbuing Agents with Long Term Memory

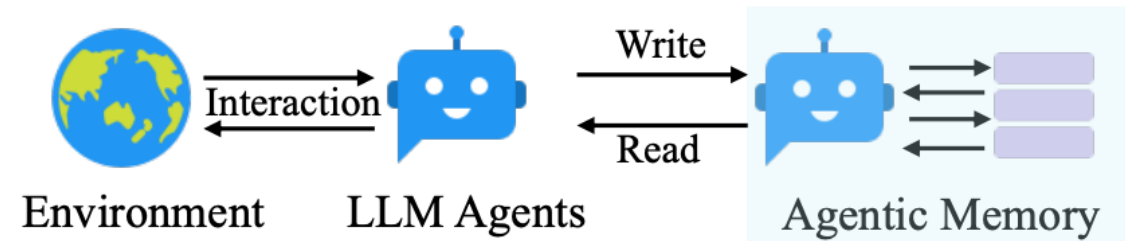


Can we model everything as an MDP?

- *Markov* implies the next state depends only on the current state and action taken.
- If the next state depends on the entire history of states, it is a partially observable MDP (POMDP)



(a) Traditional memory system.



Xu et al., A-MEM: Agentic Memory for LLM Agents:

<https://arxiv.org/pdf/2502.12110>

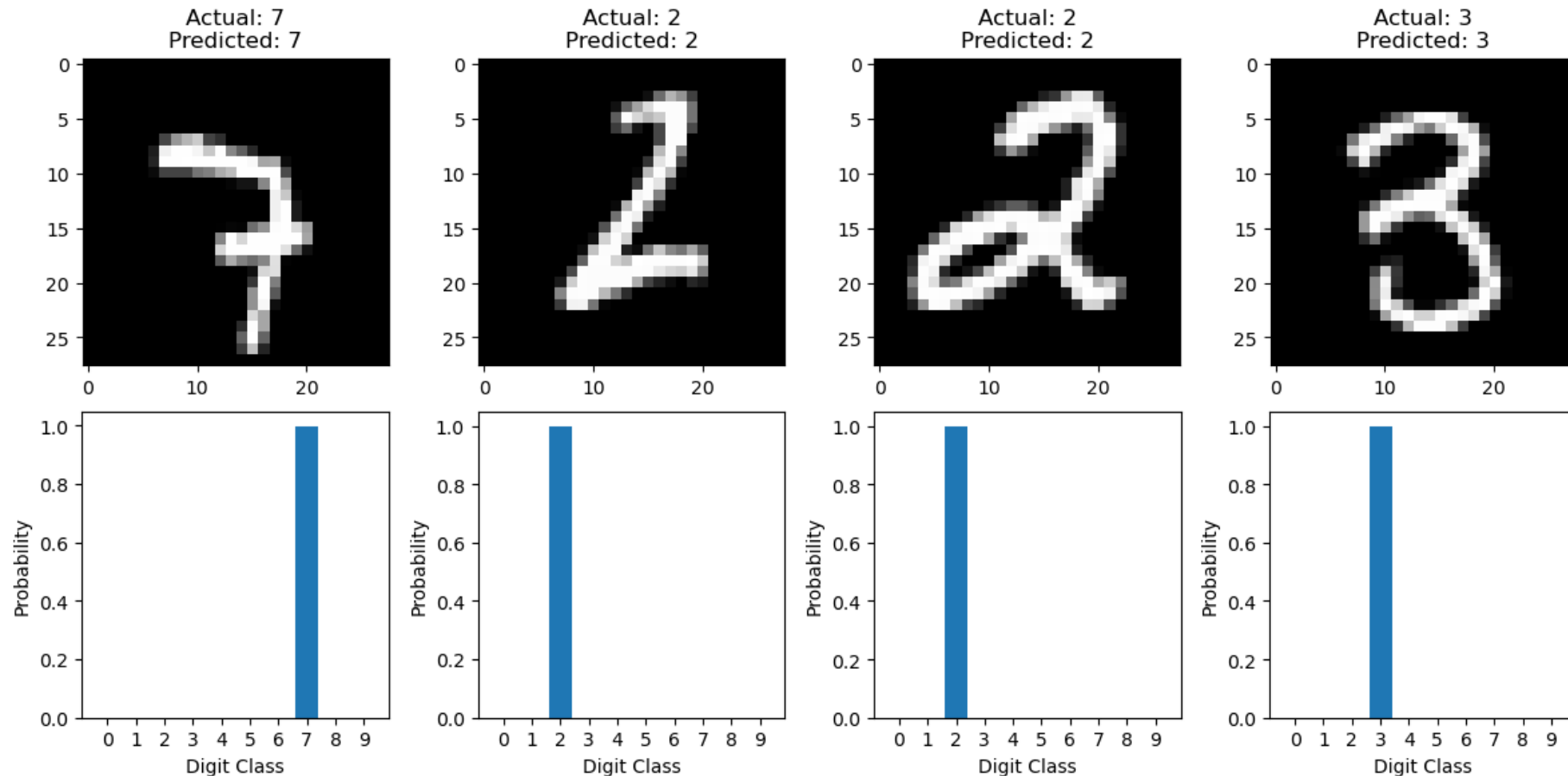
General Intelligence

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6. **Explainability and Interpretability**

Uncertainty in Deep Learning

How certain of a prediction is a Neural Network?



Uncertainty in Deep Learning

How certain of a prediction is a Neural Network?

Neural Networks can be confidently incorrect!

Why do Adversarial Attacks work?

Neural Networks are penalized for uncertainty during training



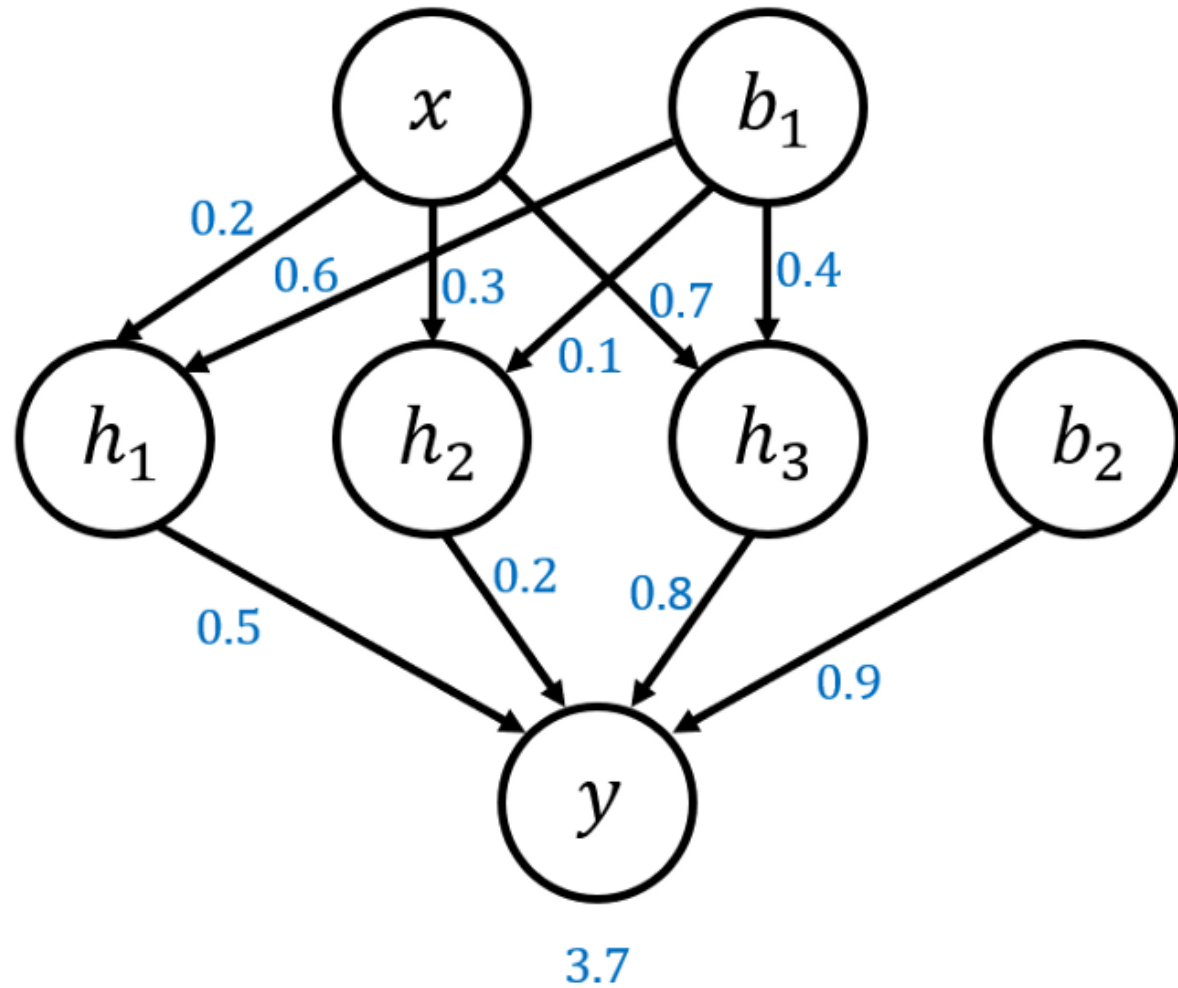
Uncertainty in Deep Learning

Help! I'm uncertain of what action to take. Please take over.

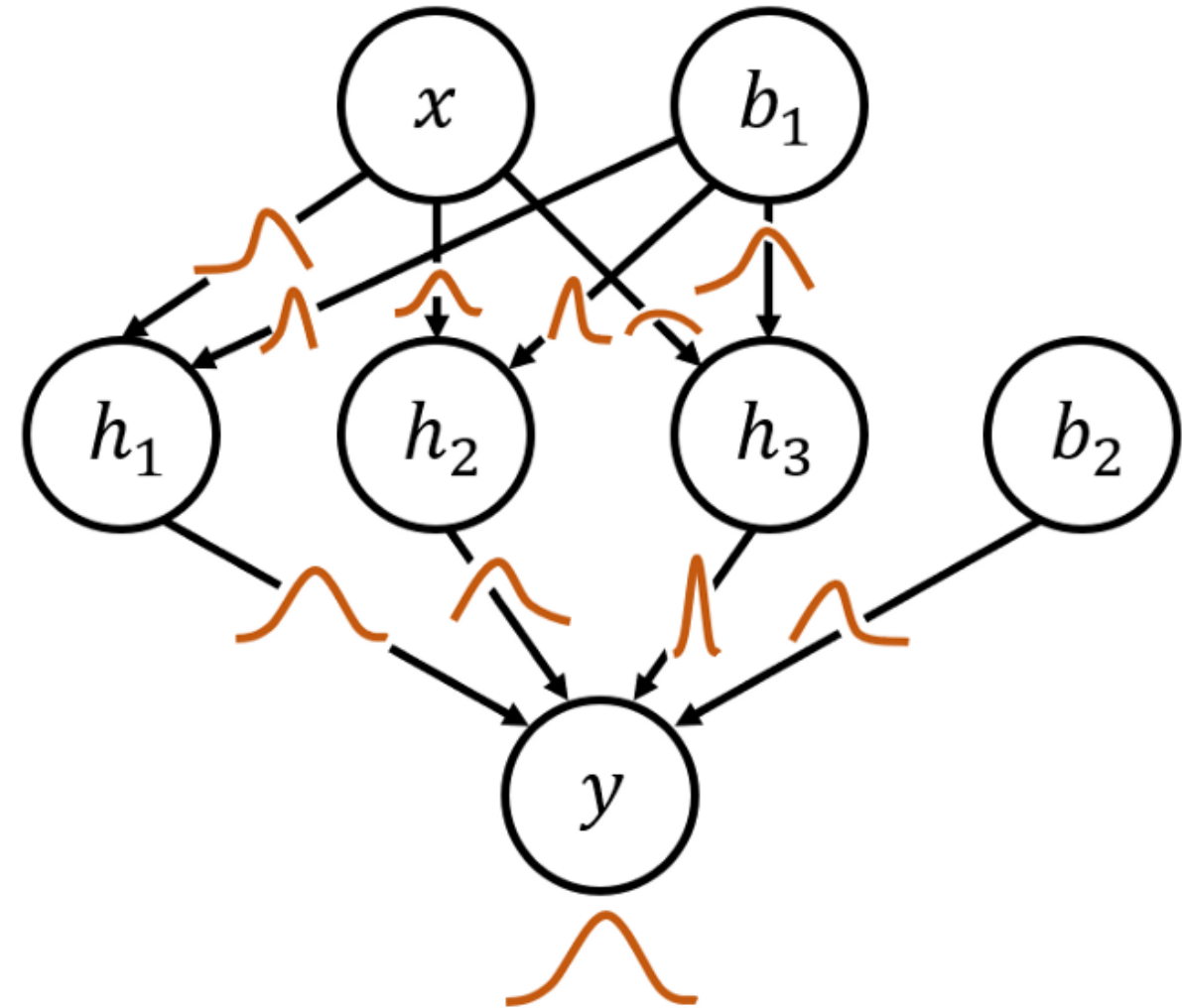
Not only want the prediction, but also an estimate of the uncertainty of that prediction



Standard Neural Network



Bayesian Neural Network



Every parameter is a distribution (i.e., $\mathcal{N}(\mu, \sigma^2)$), output is a distribution over labels, with quantified variance

Bayesian Neural Networks

The benefit of Bayesian Neural Networks is that they can model uncertainty

The downside of BNNs is that it's computationally expensive to do so...

Important Question: Are there computationally efficient ways to train neural networks to represent uncertainty?

Interpretability

LIME: What parts of an image contribute to a model's predictions?

Local Interpretable Model-agnostic Explanations (LIME)

1. Use image segmentation to group pixels together into super pixels
2. Run predictions on image with some super-pixels masked out
3. Train a simple classifier to predict which super-pixels were most important

LIME

1. Separate Image into Super-pixels
using image segmentation



Original Image



Interpretable
Components

LIME

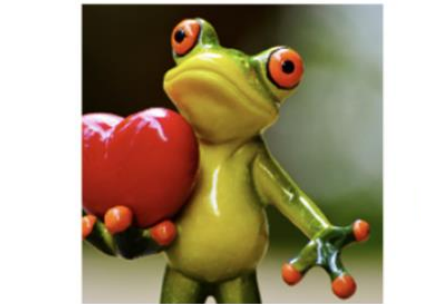
2. Run classification with some super-pixels masked

Perturbed Instances	P(tree frog)
	 0.85
	 0.00001
	 0.52

How much does the presence (or absence) of pixels affect the prediction?

LIME

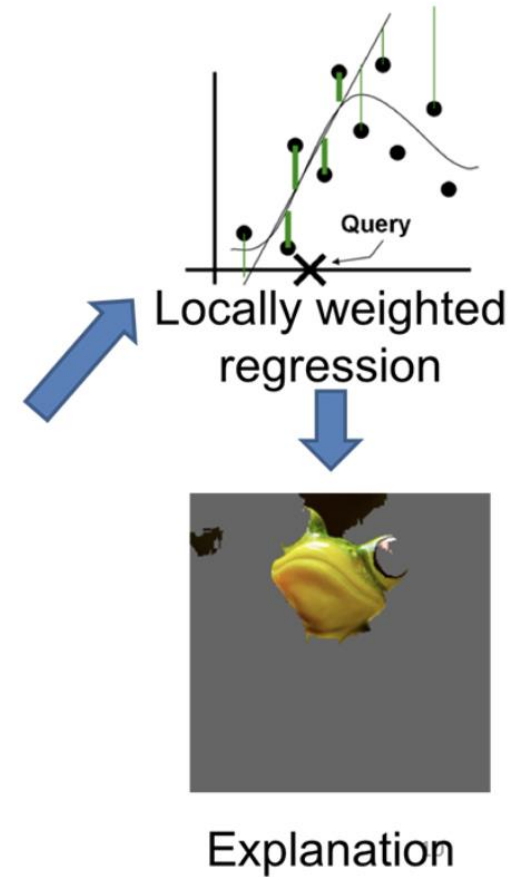
3. Train simple regression model to determine feature weighting of the super-pixels



Original Image
 $P(\text{tree frog}) = 0.54$



Perturbed Instances	$P(\text{tree frog})$
	 0.85
	 0.00001
	 0.52



Trends in Deep Learning Jobs

1. Generative Models

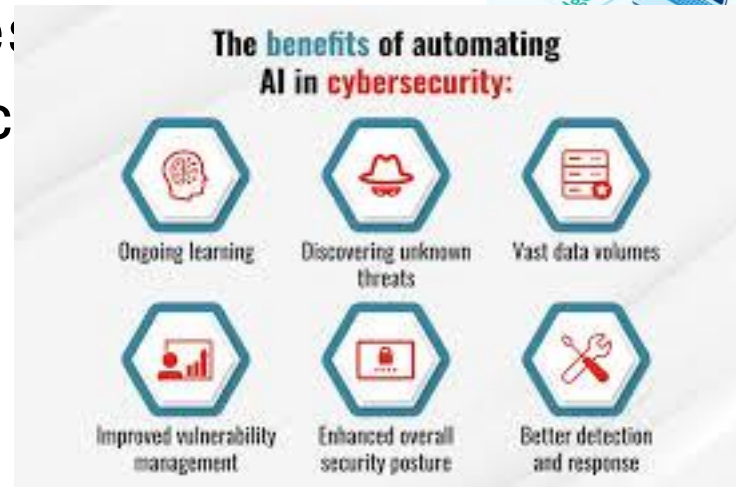


2. Edge AI, connected devices

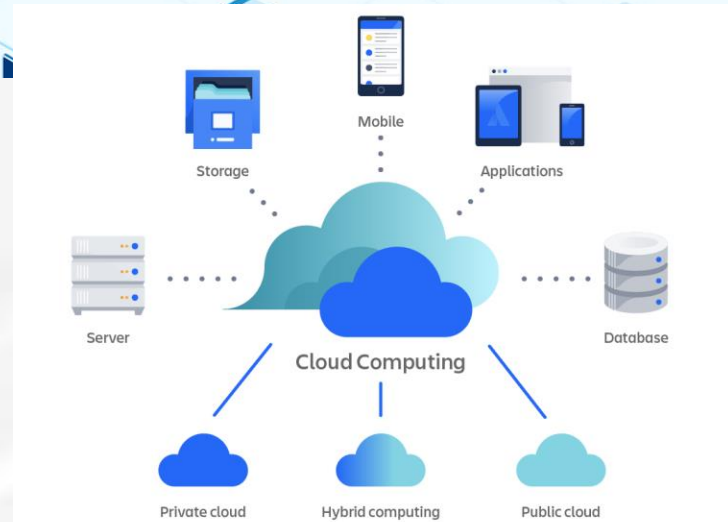
- Can your AI run on limited c

Edge AI

3. Cybersecurity and AI



4. Deep Learning Infrastructure (Hardware or Cloud Compute)



Deep Learning (or Adjacent) Jobs

- Data Scientist
 - Process and analyze large amounts of data
 - Find and validate trends in data, “Actionable Insights”
- Machine Learning Engineer
 - Build and maintain data pipeline for machine learning models
 - Implement and optimize ML/DL models
 - Evaluating Models
- Machine Learning Researcher
 - Research and implement **new** architectures, optimizers, or other techniques
- “Topic Specific” Jobs (e.g., Computer Vision, Fine-tuning LLMs, startups, etc.)

Skills You've Gained

- Foundations of linear algebra, optimization, and probability
- Data processing
- Neural network architectures
- Practical training skills (e.g., managing overfitting/underfitting, hyperparameter tuning)
- Going from math/slides to code

Skills You Should Cultivate

- Use of popular training infrastructures

The big 3 Cloud platforms



AWS



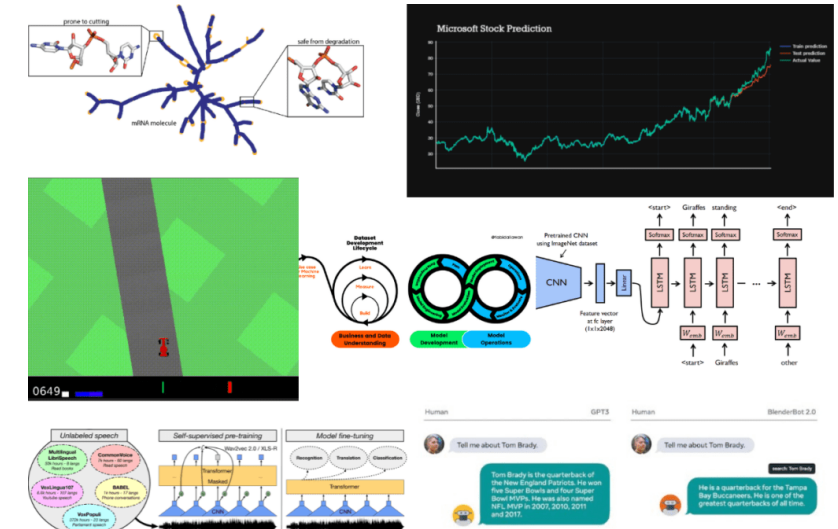
Azure



GCP

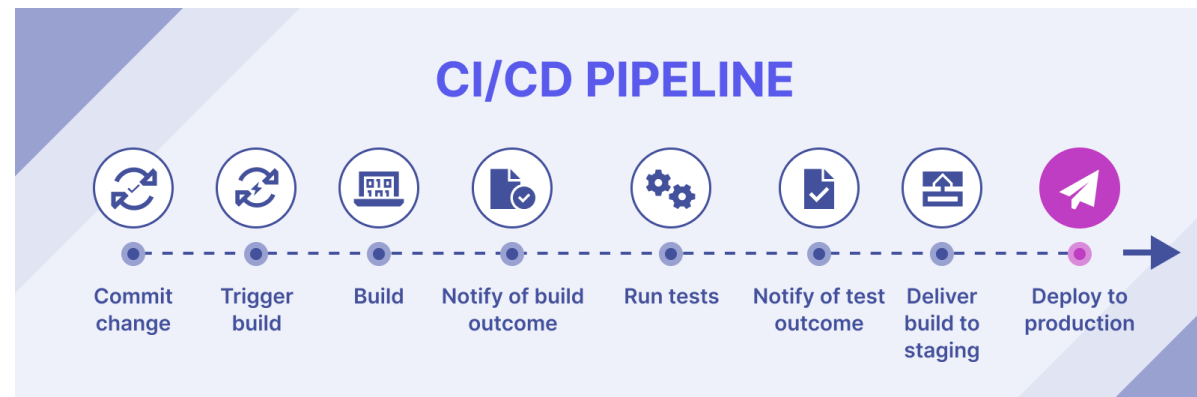
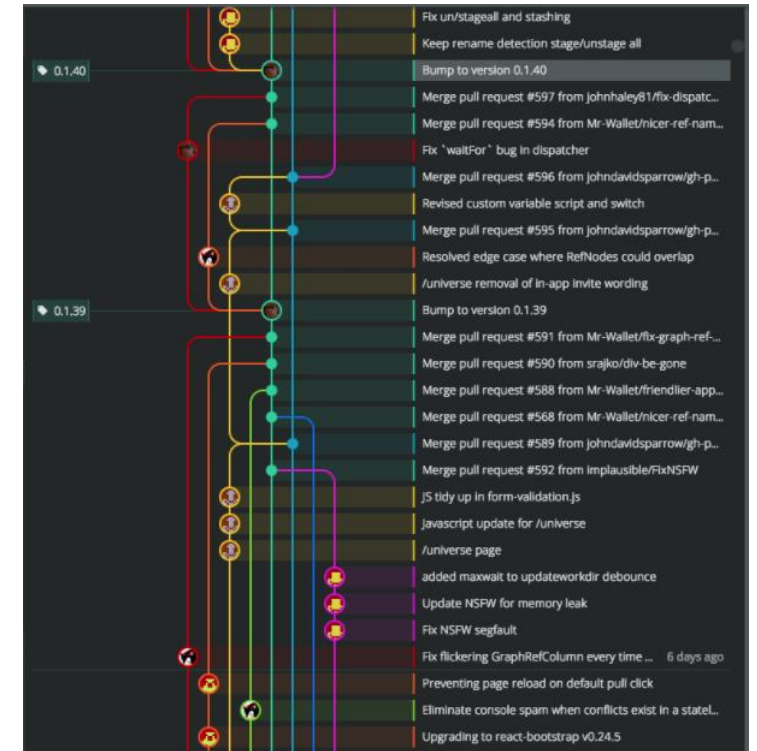


- What is the right technique to use for your problem?



Skills You Should Cultivate

- Know how to use git (more than just commit, push, and pull)
- CI/CD pipelines

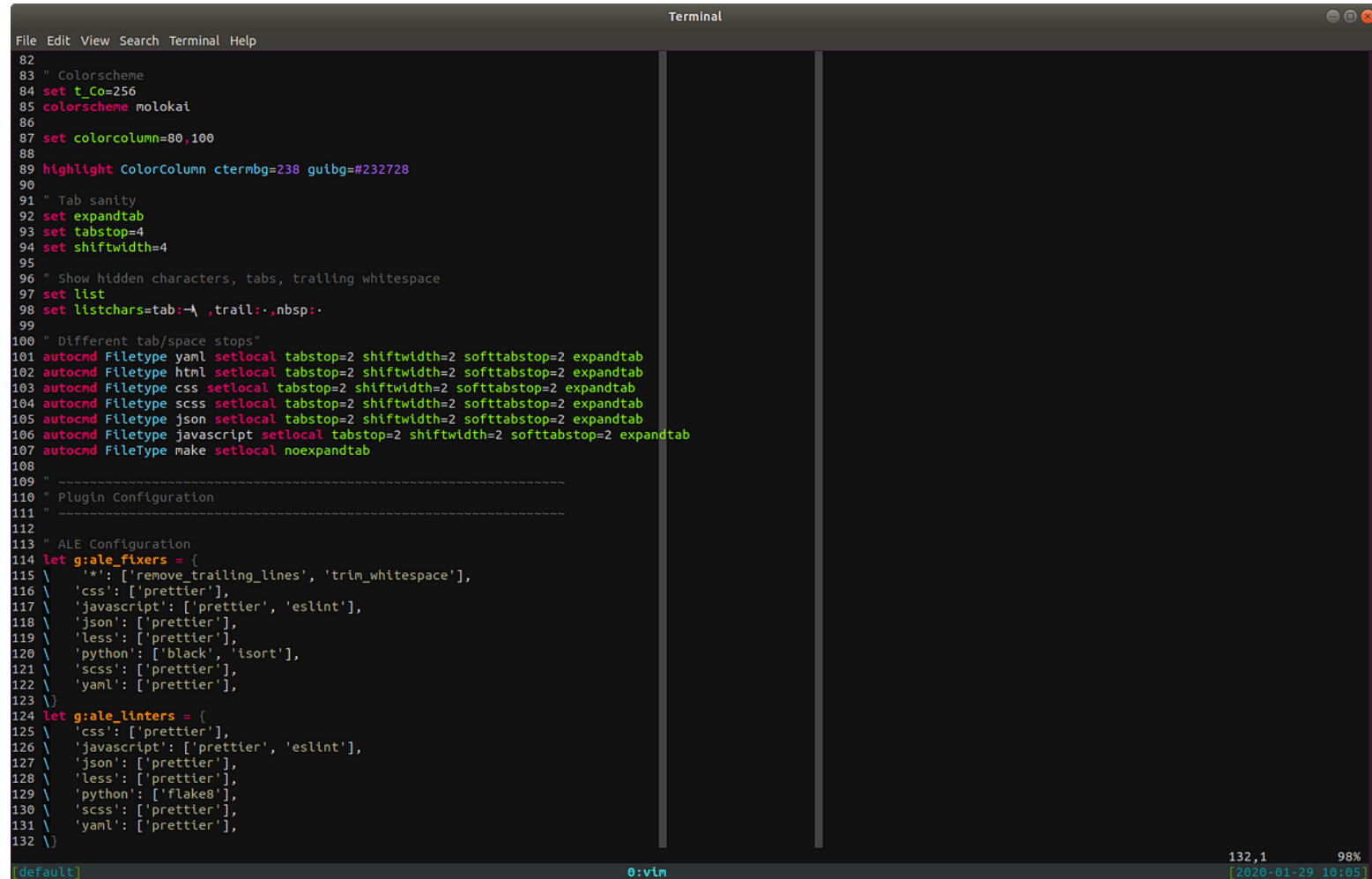


Skills You Should Cultivate

Vim

Just Kidding

(Not Really)

A screenshot of a terminal window titled "Terminal" showing a Vim configuration file (.vimrc). The file is edited in Vim, with line numbers visible on the left. The configuration includes settings for colorscheme (molokai), tab settings (expandtab, tabstop=4, shiftwidth=4), list settings (set list, listchars), autocmds for filetype settings, and ALE configuration for linting and formatting. The terminal window has a dark background with light-colored text. The status bar at the bottom shows "0:vim", "132,1", and "98%".

```
File Edit View Search Terminal Help
82
83 " Colorscheme
84 set t_Co=256
85 colorscheme molokai
86
87 set colorcolumn=80,100
88
89 highlight ColorColumn ctermbg=238 guibg=#232728
90
91 " Tab sanity
92 set expandtab
93 set tabstop=4
94 set shiftwidth=4
95
96 " Show hidden characters, tabs, trailing whitespace
97 set list
98 set listchars=tab:~\ ,trail:.,nbsp:·
99
100 " Different tab/space stops"
101 autocmd FileType yaml setlocal tabstop=2 shiftwidth=2 softtabstop=2 expandtab
102 autocmd FileType html setlocal tabstop=2 shiftwidth=2 softtabstop=2 expandtab
103 autocmd FileType css setlocal tabstop=2 shiftwidth=2 softtabstop=2 expandtab
104 autocmd FileType scss setlocal tabstop=2 shiftwidth=2 softtabstop=2 expandtab
105 autocmd FileType json setlocal tabstop=2 shiftwidth=2 softtabstop=2 expandtab
106 autocmd FileType javascript setlocal tabstop=2 shiftwidth=2 softtabstop=2 expandtab
107 autocmd FileType make setlocal noexpandtab
108
109 "
110 " Plugin Configuration
111 "
112
113 " ALE Configuration
114 let g:ale_fixers = {
115 \   '*': ['remove_trailing_lines', 'trim_whitespace'],
116 \   'css': ['prettier'],
117 \   'javascript': ['prettier', 'eslint'],
118 \   'json': ['prettier'],
119 \   'less': ['prettier'],
120 \   'python': ['black', 'isort'],
121 \   'scss': ['prettier'],
122 \   'yaml': ['prettier'],
123 \ }
124 let g:ale_linters = {
125 \   'css': ['prettier'],
126 \   'javascript': ['prettier', 'eslint'],
127 \   'json': ['prettier'],
128 \   'less': ['prettier'],
129 \   'python': ['flake8'],
130 \   'scss': ['prettier'],
131 \   'yaml': ['prettier'],
132 \ }
[default] 0:vim 132,1 98% [2020-01-29 10:05]
```

Build a Portfolio

- Portfolios matter!
 - Demonstrates practical skills beyond theoretical knowledge
 - Shows ability to complete projects from conception to deployment
 - Acts as tangible proof of your capabilities
- Essential Portfolio Components
 - GitHub repository with well-documented code
 - Personal website showcasing projects
 - Variety of projects demonstrating range of skills
- Ideally shows a mix of breadth in many topics and depth in a certain topic

What Type of Projects to do?

- A bit easier to say what not to do:
 - Don't make another CNN for MNIST or CIFAR
 - Don't find a popular reddit post or medium article describing "Top 5 projects for your DL Portfolio"
- Projects are evaluated for creativity first and then technicality
 - The first person who looks at your portfolio won't be able to evaluate your technical ability, you must draw them in
- A portfolio doesn't have to be perfectly cohesive
 - Follow your short-term nose

What Type of Projects to do?

- (At least part of) your portfolio should match the jobs you are applying for
 - If you are applying for a ML-Engineer or data science position, you **shouldn't** work only with perfectly clean data (e.g., Kaggle datasets)
- Do what interests **you**
 - You will work harder and create better results if you are **genuinely interested** in your project
- If you're interested in grad school, get involved in research
 - If you're in grad school, try doing research

Research Needs a Special Sauce

- There are lots of people doing research in AI
 - Be wary of problems you know thousands of people are working on
 - Don't try to make a better CNN for ImageNet
- Research needs something special:
 - A technique that other people haven't thought of applying to this problem yet
 - Harder, but more impactful
 - A dataset that other people don't have access to
 - Easier, but potentially isolated impact

Should I Use LLMs to Build my Portfolio?



Should I Use LLMs to Build my Portfolio?

The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers

Hao-Ping (Hank) Lee
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
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Microsoft Research
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serintel@microsoft.com

Nicholas Wilson
Microsoft Research
Cambridge, United Kingdom
niwilson@microsoft.com

Lev Tankelevitch
Microsoft Research
Cambridge, United Kingdom
levt@microsoft.com

Richard Banks
Microsoft Research Cambridge
Cambridge, United Kingdom
rbanks@microsoft.com

Should I Use LLMs to Build my Portfolio?

My Take: If you want a job in ML, ***you*** should do the machine learning

If you want a job as a ML engineer, you shouldn't only work with perfectly clean data

If you want to showoff your ML project as part of a webapp, it's reasonable to automate building the frontend.

Any Questions?